GAZE — Data Card

Pointing04 moderate-pose subset · CP2 hand-in · 22 October 2026

Source and permission

The head-direction dataset is a selected subset of the author-hosted Pointing04 Head Pose Image Database by Gourier, Hall, and Crowley (2004). The author permits reuse for any purpose with source attribution; this is a custom attribution condition. The source has 15 people and two pose series per person. Full data stays outside Git; a reproducible download and selection script is included. Only 20 small permitted examples are committed.

Author source and permission | Original archive

Prepared split and class counts

Split	People	Forward	Left	Right	Down	Total
Train: not applicable	0	0	0	0	0	0
Validation	10	60	60	60	60	240
Reserved test	5	30	30	30	30	120
Demo examples*	5	5	5	5	5	20

*The 20 examples are part of validation, not additional data. The selected dataset totals 360 unique images. People and recording sources remain within one split; validation/test subject, source, ID, path, and pixel-hash intersections are empty. No project-specific classifier is trained: pretrained landmark inference feeds an untrained geometric rule. Validation is used for settings; reserved-test results have not been used for tuning.

Labels and checks

Forward, left, right, and down describe the person's direction in unmirrored input. Labels come from nominal source pose annotations, not predictions. Source angle signs and representative cardinal poses were visually checked. All selected files were decoded and their dimensions verified. Missing or ambiguous predictions remain unknown and count as unsuccessful classifications in overall accuracy.

Image sizes and preprocessing

All selected images are 384 × 288 color JPEGs. Shared prediction/evaluation functions convert decoded color channels for the model and shrink oversized images to a maximum width of 1280 while preserving proportions. These selected images are not resized. No fitted preprocessing, project training, or augmentation is used. Before/after examples document the applied preparation.

Coverage and known gaps

This moderate-angle, single-person posed subset excludes upward-only views, extreme profiles, and many mixed poses. It cannot establish performance on distant/occluded classroom faces, natural writing, seat association, or movement events. A separate public-domain FourPeople office video from Xiph.Org Foundation supplies a multi-person pipeline proxy; it has no independently completed classroom/event labels.

Acquisition and reproducibility

Run scripts/prepare_data.py to retrieve the source archive, verify its pinned byte checksum, and regenerate the fixed manifests. Full downloaded images remain outside Git. The detailed repository data card records selection rules, source conditions, filenames, and integrity values.

References

Gourier, N., Hall, D., & Crowley, J. L. (2004). *Estimating face orientation from robust detection of salient facial structures* [Workshop paper]. Pointing '04, Cambridge, United Kingdom. https://homepages.inf.ed.ac.uk/rbf/CAVIAR/PAPERS/Pointing04-Gourier.pdf

Xiph.Org Foundation. (n.d.). *Xiph.org video test media [derf's collection]* [Video collection]. https://media.xiph.org/video/derf/